

# Midterm 1

ORF 309, Fall 2025

**Instructions:** This midterm has 3 questions for a total of 90 points. Partial credit will be awarded for partial proofs or arguments. Strive to reduce answers as much as possible (e.g., give a fraction in simplest terms, though a numerical answer requiring a calculator is not necessary). You may freely use basic facts from class, such as laws of probability, properties of conditional expectation, or manipulations with pmfs/pdfs. However, you must provide adequate justifications for important steps. Put another way, try your best to demonstrate awareness that you are applying the correct facts in the correct way by explicitly mentioning the results you are applying, indicating the objects they are applied to, and checking the required assumptions.

**Length:** 75 minutes (for working on the problems) + 15 minutes (for submission)

Good Luck!

| Question | Points | Score |
|----------|--------|-------|
| 1        | 30     |       |
| 2        | 30     |       |
| 3        | 30     |       |
| Total:   | 90     |       |

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1. A factory produces widgets on two assembly lines, Line A and Line B.
    - Line A produces 60% of the total widgets and has a defect rate of 5%.
    - Line B produces the remaining 40% of widgets and has a defect rate of 10%.

All widgets are sent to a central quality control station for testing.

- (a) (10 points) What is the probability that a randomly selected widget is defective?
- (b) (10 points) You are handed a widget that has been identified as defective. What is the probability it came from Line A?
- (c) (10 points) A customer receives a shipment of 10 widgets, selected at random from the entire production. What is the probability that this shipment contains exactly 3 defective widgets?

2. ChatGPT (and similar large language models) is still imperfect at solving hard math problems. What further complicates matters is that even with the same fixed prompt, its outputs can vary randomly. Suppose that for a given math problem, ChatGPT's responses from repeated trials are independent and uniformly distributed over the set  $\{0, 100, 200, 300\}$ . Let  $Z_n$  denote the average of the first  $n$  responses, and  $F_n$  denote the fraction of 300's among them.
- (a) (10 points) Compute the expectation of  $Z_n$ .
  - (b) (10 points) Compute the variance of  $Z_n$ .
  - (c) (10 points) What can you say about  $F_n$ , if  $n$  is a very large number? What about the probability of the event  $\{Z_n = 300\}$ ?

3. Alice and Bob, two arcade enthusiasts, play a claw machine until each wins their first prize. The probability of winning a prize on any single attempt is  $p = 0.2$ . Each attempt is independent and costs one dollar.
- (a) (10 points) What is the probability that Alice wins her first prize with exactly 3 attempts, given that her first attempt failed?
  - (b) (10 points) Given that Bob's first attempt failed, what is his expected total cost to win his first prize?
  - (c) (10 points) On average, how much does the more unlucky player (the one who spends more) expect to pay? In other words, if  $N_A$  and  $N_B$  denote the total numbers of attempts made by Alice and Bob until their first win, find

$$\mathbb{E}[\max(N_A, N_B)].$$